

Masked Hypertension and Left Atrial Dysfunction: A Hidden Association

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Masked hypertension (MH) is a clinical condition that indicates normal values of clinic blood pressure (BP) but elevated 24-hour BP. The purpose of this study was to investigate the relationship between MH and left atrial (LA) phasic function evaluated by both the volumetric and speckle tracking method. This cross-sectional study included 49 normotensive individuals, 50 patients with MH, and 70 untreated sustained hypertensive patients adjusted by age and sex. MH was diagnosed if clinic BP was normal and 24-hour BP was increased. LA reservoir function was lower in patients with MH and those with sustained hypertension compared with the normotensive group. LA conduit

function gradually decreased, while LA booster pump function progressively increased, from normotension to sustained hypertension. Similar results were obtained by two-dimensional echocardiographic strain analysis. Independently of main clinic and echocardiographic characteristics, 24-hour systolic BP was associated with LA passive ejection fraction, LA total longitudinal strain, LA positive longitudinal strain, and LA stiffness index. In conclusion, MH is associated with impairment of LA phasic function and stiffness, and 24-hour systolic BP increment was closely related with LA remodeling. *J Clin Hypertens (Greenwich)*. 2017;19:305–311. ©2016 Wiley Periodicals, Inc.

At the beginning of the new millennium, the increasing use of ambulatory blood pressure (BP) monitoring (ABPM) has revealed the whole range of different BP patterns such as white-coat hypertension, sustained hypertension, nocturnal hypertension, and normotension. This was the reason for the introduction of 24-hour ABPM in guidelines regarding diagnosis of arterial hypertension and in everyday clinical practice. ABPM could provide true and prognostic information on BP variability in hypertensive patients.¹ Recent analysis shows important prognostic value of BP variability, assessed by 24-hour ABPM, in white-coat hypertension among older patients.²

Thomas Pickering was the first to use the term *masked hypertension* (MH).³ Shortly thereafter investigations appeared that demonstrated a negative influence of MH on target organ damage, including cardiac hypertrophy, carotid atherosclerosis, and albuminuria.^{4–6} These changes could explain the unfavorable prognosis of patients with MH and increased cardiovascular morbidity and mortality in this population.^{7,8} Our recent meta-analysis included 776 patients with MH and showed that MH is associated with significant risk of left ventricular (LV) structural changes, primarily LV hypertrophy.⁹

Left atrial (LA) remodeling has not been sufficiently investigated in MH so far. Sincer and colleagues¹⁰ provided data only on LA diameter, without any further

details regarding LA volume (LAV) and phasic function, and they did not find a significant difference between normotensive control, sustained hypertensive, and MH patients.¹⁰ In addition, LA remodeling could significantly contribute to increased morbidity and mortality in MH. In this context, recent findings by Gupta and colleagues¹¹ and Patel and colleagues¹² showed that LA enlargement, as well as LV dysfunction and LV impaired mechanics, was associated with adverse outcomes in the general population. Thus, it appeared that the comprehensive assessment of LA phasic function may be of clinical importance in MH patients and might be helpful in the risk stratification of these patients.

We hypothesized that MH affects LA function and mechanics. The aim of the present study was to investigate LA phasic function in patients with MH and sustained hypertension vs those with normotension using volumetric and speckle tracking analysis.

METHODOLOGY

This investigation included 169 untreated patients referred to our outpatient clinic for ABPM. Arterial hypertension was diagnosed according to current guidelines.¹³ Exclusion criteria were antihypertensive treatment, symptoms or signs of heart failure, coronary artery disease, previous cerebrovascular events, atrial fibrillation, congenital heart disease, heart valve disease (more than mild), neoplastic disease, liver cirrhosis, kidney failure, and endocrine diseases (including type 2 diabetes mellitus).

Anthropometric measures (height, weight) and laboratory analyses (level of fasting glucose, blood creatinine, total cholesterol, and triglycerides) were obtained from all of the patients included in the study. Body mass index (BMI) and body surface area (BSA) were calculated for each patient.

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The study was approved by the local ethics committee, and informed consent was obtained from all participants.

Clinic BP Measurement and 24-Hour ABPM

All participants underwent 24-hour ABPM according to current guidelines.¹³ Clinic arterial BP values were obtained by aneroid manometer in the morning hours by calculating the average value of the three consecutive measurements in the sitting position, taken within an interval of 5 to 10 minutes, after the patient had rested for at least 5 minutes in that position.

The noninvasive 24-hour ABPM was performed on the nondominant arm, using a Schiller BR-102 plus system BP monitor (Schiller AG, Baar, Switzerland). The device was programmed to obtain BP readings at 20-minute intervals during the day (7 AM–11 PM) and at 30-minute intervals during the night (11 PM–7 AM). Nighttime BP was defined as the average of BPs from the time patients went to bed until the time they got out of bed, and daytime BP was defined as the average of BPs recorded during the rest of the day. The recording was then analyzed to obtain 24-hour average systolic BP (SBP) and diastolic BP (DBP). When the readings exceeded at least 70% of the total readings programmed for the testing period, the recording was considered valid and satisfactory.

MH was defined as a normal clinic BP (SBP <140 mm Hg and DBP <90 mm Hg) measured on at least two separate occasions associated with a 24-hour ambulatory SBP >130 mm Hg and/or DBP >80 mm Hg. On the other hand, patients with sustained hypertension had an elevated clinic SBP of 140 mm Hg together with a 24-hour ambulatory SBP ≥130 mm Hg or DBP ≥80 mm Hg.

Echocardiography

Echocardiographic examinations were performed using a commercially available Vivid 7 device (GE Vingmed, Horten, Norway). Reported values of all two-dimensional echocardiographic (2DE) parameters were obtained as the average value of three consecutive cardiac cycles. LV dimensions were measured according to joined guidelines of the American and European societies of echocardiography.¹⁴ Relative wall thickness was calculated according to the formula: $(2 \times \text{posterior wall thickness})/\text{LV end-diastolic diameter}$. Posterior wall thickness was measured at end-diastole. LV ejection fraction (EF) was calculated by using the biplane method. LV mass (LVM) was calculated by using the corrected method described in the joined guidelines of the American and European societies of echocardiography and indexed for BSA.¹⁴

Doppler assessment of transmitral flow was obtained in the apical four-chamber view according to the guidelines.¹⁵ Tissue Doppler imaging was used to obtain LV myocardial velocities in the apical four-chamber view, with a sample volume placed at the septal and lateral segments of the mitral annulus during early

diastole (e'). The average of the peak early diastolic relaxation velocity (e') of the septal and lateral mitral annulus was calculated, and the E/e' ratio was computed.

2DE ASSESSMENT OF LV VOLUME AND FUNCTION

LAVs were measured in three different sequences of the cardiac cycle: maximal LAV was measured just before the mitral valve opening, pre-A (pre atrial contraction) LAV was determined at the onset of atrial systole (peak of P wave in ECG), and minimal LAV was measured at the mitral valve closure. All volumes were determined according to the biplane method in four- and two-chamber views, and all the values were indexed for BSA. The total emptying volume – parameter of LA reservoir function was calculated as the difference between maximum and minimum LAV, passive emptying volume – conduit function parameter was calculated as the difference between maximum and pre-A LAV, and active emptying volume – LA booster function parameter was obtained as the difference between pre-A and minimum LAV.¹⁶ Consequently, total emptying fraction (EF) was calculated as the ratio between total emptying volume and maximum LAV, passive EF was computed as the ratio between passive and maximum, and active EF was obtained as the proportion between active and pre-A LAV.

2DE strain imaging was performed in the apical four- and two-chamber views¹⁶ and a commercially available software Echo PAC 201 (GE Healthcare, Horten, Norway) was used for the 2DE strain analysis. LA strain was calculated from the peak of the P wave to the peak of the P wave in the next cardiac cycle. The LA endocardium was manually traced. An average longitudinal strain curve was automatically generated, and it included a negative deflection (LA negative longitudinal strain), representing LA active contraction, followed by a positive one during LA filling (LA positive longitudinal strain). Their summation represented total LA longitudinal strain. LA strain (positive, negative, and total) was calculated by averaging the values obtained in four- and two-chamber apical views.

LA stiffness index was evaluated as the ratio between mitral E/e' (e' is the average between septal and lateral e' values) and LA total longitudinal strain.

Statistical Analysis

Continuous variables were presented as mean ± standard deviation and were compared by analysis of variance, as they showed normal distribution. Bonferroni post hoc analysis was used for the comparison between different groups. Differences in proportions were compared by the χ^2 test. Multivariate regression analysis was used to determine the association between different echocardiographic parameters and SBP (clinic and 24-hour) independently of age, BMI, LVM index (LVMI), and E/e' . A P value <.05 was considered statistically significant.

RESULTS

All three groups were of similar age and sex distribution (Table I). There was no difference in BMI, although BSA was somewhat higher in the MH and sustained hypertensive groups than in controls (Table I). Heart rate was similar between all groups. Clinic BP was similar between controls and the MH group, and significantly higher in the sustained hypertensive group (Table I). On the other hand, 24-hour BP was significantly higher in the MH and sustained hypertensive groups than in controls (Table I). Plasma glucose and creatinine levels were similar throughout the groups, whereas cholesterol and triglyceride levels were higher in the MH and hypertensive groups.

Echocardiographic Measurements

LV diameters were similar among the groups, while LV interventricular, posterior wall, and relative wall thicknesses were significantly higher in the MH and sustained hypertensive patients than in controls (Table II). LVMI progressively increased from normotensive across MH patients to hypertensive patients (Table II). There was no difference in LV 2DE EF between all three groups. Mitral E/A ratio decreased and mitral E/e' ratio increased gradually from normotensive to sustained hypertensive (Table II).

LA Phasic Function

All LAVs (maximum, minimum, and pre-A) and corresponding LAV indexes progressively and significantly increased from normotensive controls to sustained hypertensive patients (Table III).

There was no significant difference in total LAV between groups. Total LA EF, which describes LA reservoir function, was lower in MH and sustained hypertensive patients than in normotensive controls (Table III and Figure). Passive LAV was lower in hypertensive patients than in controls and MH patients, and passive LA EF, a parameter of LA conduit function,

gradually declined from normotensive controls, throughout MH, to sustained hypertensive patients (Table III and Figure). Active LAV and active LA EF gradually increased from normotensives to sustained hypertensive patients (Table III and Figure).

Total and positive LA longitudinal strain gradually decreased from controls to sustained hypertensive patients, whereas negative LA longitudinal strain was higher in MH and sustained hypertensives than in normotensive controls (Table III and Figure).

LA stiffness index significantly and progressively increased from normotensive controls, across MH, to sustained hypertensive patients (Table III).

Regression Analysis

Clinic SBP was associated with LA passive EF ($\beta=-0.253$, $P=.012$), LA total longitudinal strain ($\beta=-0.308$, $P<.001$), and LA stiffness index ($\beta=0.382$, $P<.001$) independently of age, BMI, LV structure, and diastolic function in the whole study population.

Twenty-four-hour SBP was associated with LA passive EF ($\beta=-0.331$, $P<.001$), LA total longitudinal strain ($\beta=-0.376$, $P<.001$), LA positive longitudinal strain ($\beta=-0.307$, $P<.001$) and LA stiffness index ($\beta=0.437$, $P<.001$) independently of main clinical and echocardiographic characteristics in the whole study population.

DISCUSSION

Our study revealed several new findings. First, maximum, minimum, and pre-A LAVs were significantly higher in MH patients than in controls, but still lower than in sustained hypertensive patients. Second, conduit and reservoir LA function were reduced in MH patients compared with controls, whereas LA pump function increased from controls to sustained hypertensive patients. Third, LA stiffness index progressively increased from normotensives, throughout MH, to sustained hypertensive patients. Fourth, 24-hour SBP

TABLE I. Demographic Characteristics and Clinical Parameters of the Study Population

	Normal BP (n=49)	MH (n=50)	HTN (n=70)	P Value
Age, y	56±7	59±7	58±7	.097
Women, %	23 (47)	22 (44)	32 (46)	.957
BMI, kg/m ²	25.3±2.4	26.2±2.3	26.1±2.2	.096
BSA, m ²	1.91±0.13	1.97±0.14 ^a	1.97±0.13 ^a	.031
Heart rate, beats per min	71±6	71±7	72±8	.672
Clinic systolic BP, mm Hg	127±7	131±8	152±11 ^{b,c}	<.001
Clinic diastolic BP, mm Hg	72±7	74±6	92±8 ^{b,c}	<.001
24-h systolic BP, mm Hg	118±8	139±8 ^b	142±9 ^b	<.001
24-h diastolic BP, mm Hg	69±6	85±7 ^b	87±7 ^b	<.001
Plasma glucose, mmol/L	5.1±0.7	5.3±0.8	5.4±0.7 ^a	.090
Triglycerides, mmol/L	1.8±0.3	2.1±0.3	2.3±0.4	<.001 ^[**]
Total cholesterol, mmol/L	5.5±0.6	5.9±0.7 ^a	6.0±0.7 ^b	<.001
Serum creatinine, umol/L	79±9	78±8	81±9	.159

Abbreviation: BMI, body mass index; BP, blood pressure; BSA, body surface area; HTN, hypertension; MH, masked hypertension. ^a $P<.05$ vs normal BP.

^b $P<.01$ vs normal BP. ^c $P<.01$ vs MH. ^[**] $P<.01$ for all comparisons.

TABLE II. Echocardiographic Parameters of Left Ventricular Structure and Function in the Study Population

	Normal BP (n=49)	MH (n=50)	HTN (n=70)	P Value
LVEDD, mm	48.3±4.1	49.2±4.2	49.7±4.3	.206
LVESD, mm	29.6±3.2	30.1±3.1	30.7±3.4	.189
IVS, mm	9.2±1.0	10.2±1.2 ^a	10.5±1.3 ^a	<.001
PWT, mm	9.1±1.1	10.1±1.1 ^a	10.3±1.2 ^a	<.001
RWT	0.38±0.03	0.41±0.03 ^a	0.41±0.03 ^a	<.001
LVMI, g/m ²	81±8	91±10	96±12	<.001 ^b
EF, %	63±3	63±3	62±3	.106
E/A ratio	1.17±0.24	0.93±0.20 ^a	0.84±0.18 ^a	<.001
E/e'	6.2±1.9	8.1±2.0	9.4±2.3	<.001 ^c

Abbreviations: A, late diastolic mitral flow (pulse Doppler); e', average of the peak early diastolic relaxation velocity of the septal and lateral mitral annulus (tissue Doppler); E, early diastolic mitral flow (pulsed Doppler); EF, ejection fraction; HTN, hypertension; IVS, interventricular septum; LVEDD, left ventricular end-diastolic dimension; LVESD, left ventricular end-systolic dimension; LVMI, left ventricular mass index; MH, masked hypertension; PWT, posterior wall thickness; RWT, relative wall thickness. ^aP<.01 vs normal blood pressure. ^bP<.05 for all comparisons. ^cP<.01 for all comparisons.

TABLE III. LA Phasic Function Assessed by Volumetric and Strain Methods in the Study Population (Strain Parameters Adjusted for BMI)

	Normal BP (n=49)	MH (n=50)	HTN (n=70)	P Value
LAV analysis				
LAV _{max} , mL	56.0±7.0	62.3±7.6 ^a	65.6±8.2 ^a	<.001
LAV _{max} /BSA, mL/m ²	29.3±3.6	31.6±4.1 ^b	33.3±4.2 ^a	<.001
LAV _{min} , mL	23.0±3.3	27.4±3.9	30.2±4.4	<.001 ^c
LAV _{min} /BSA, mL/m ²	12.0±2.2	13.9±2.4	15.3±3.3	<.001 ^d
LAV _{pre-a} , mL	32.9±5.4	40.9±6.3	46.5±7.8	<.001 ^c
LAV _{pre-a} /BSA, mL/m ²	17.2±3.0	20.8±3.4	23.6±4.1	<.001 ^c
LA TotEV, mL	33.0±5.4	34.9±6.0	35.4±6.4	.092
LA PassEV, mL	23.1±4.4	21.5±3.8	19.0±3.9 ^{a,e}	<.001
LA ActEV, mL	9.9±2.9	13.5±3.7	16.3±4.2	<.001 ^c
LA TotEF, %	59±6	56±4 ^b	54±5 ^a	<.001
LA PassEF, %	41±5	34±5	29±4	<.001 ^c
LA ActEF, %	30±4	33±4	35±5	<.001 ^d
LA mechanics				
Total longitudinal strain, %	39±5	35±4	33±4	<.001 ^d
Positive longitudinal strain, %	24±5	17±4	13±3	<.001 ^c
Negative longitudinal strain, %	-15±4	-18±5 ^b	-20±6 ^a	<.001
LA stiffness index	0.16±0.05	0.23±0.06	0.28±0.08	<.001 ^c

Abbreviation: ActEF, active emptying fraction; ActEV, active emptying volume; BMI, body mass index; BP, blood pressure; HTN, hypertension; LA, left atrial; LAV, left atrial volume; LAV_{max}, maximal left atrial volume; LAV_{min}, minimal left atrial volume; LAV_{pre-a}, left atrial volume before atrial contraction; PassEF, passive emptying fraction; PassEV, passive emptying volume; TotEF, total emptying fraction; TotEV, total emptying volume; WCH, white-coat hypertension. ^aP<.01 vs WCH. ^bP<.01 vs normal BP. ^cP<.01 for all comparisons. ^dP<.05 for all comparisons. ^eP<.01 vs MH.

better correlated with LA functional and mechanical parameters than clinic SBP in the whole study population.

There is no consistency in findings related with LA phasic function in the hypertensive population. Some authors have found that all LA functions are reduced in hypertensive patients,¹⁷ whereas other investigators reported decreased conduit and enhanced LA reservoir and pump function.¹⁸ In addition, other researchers have demonstrated reduced LA reservoir and conduit but compensatory increased booster pump LA function.¹⁹ Our results concur with the last group of authors. However, we went one step further in

investigating MH patients and revealed that LA reservoir and conduit function, assessed by volumetric and strain analysis, were impaired in these patients. On the other hand, LA booster pump function was enhanced in the MH group. Interestingly, although LA function was obviously impaired in MH participants compared with controls, the significant difference still existed between MH and sustained hypertensive patients. This might potentially be explained by the different duration of hypertension (masked or sustained) in our groups. Considering the fact that we included only untreated patients, we could not report the duration of hypertension, but it is possible that sustained hypertension lasted

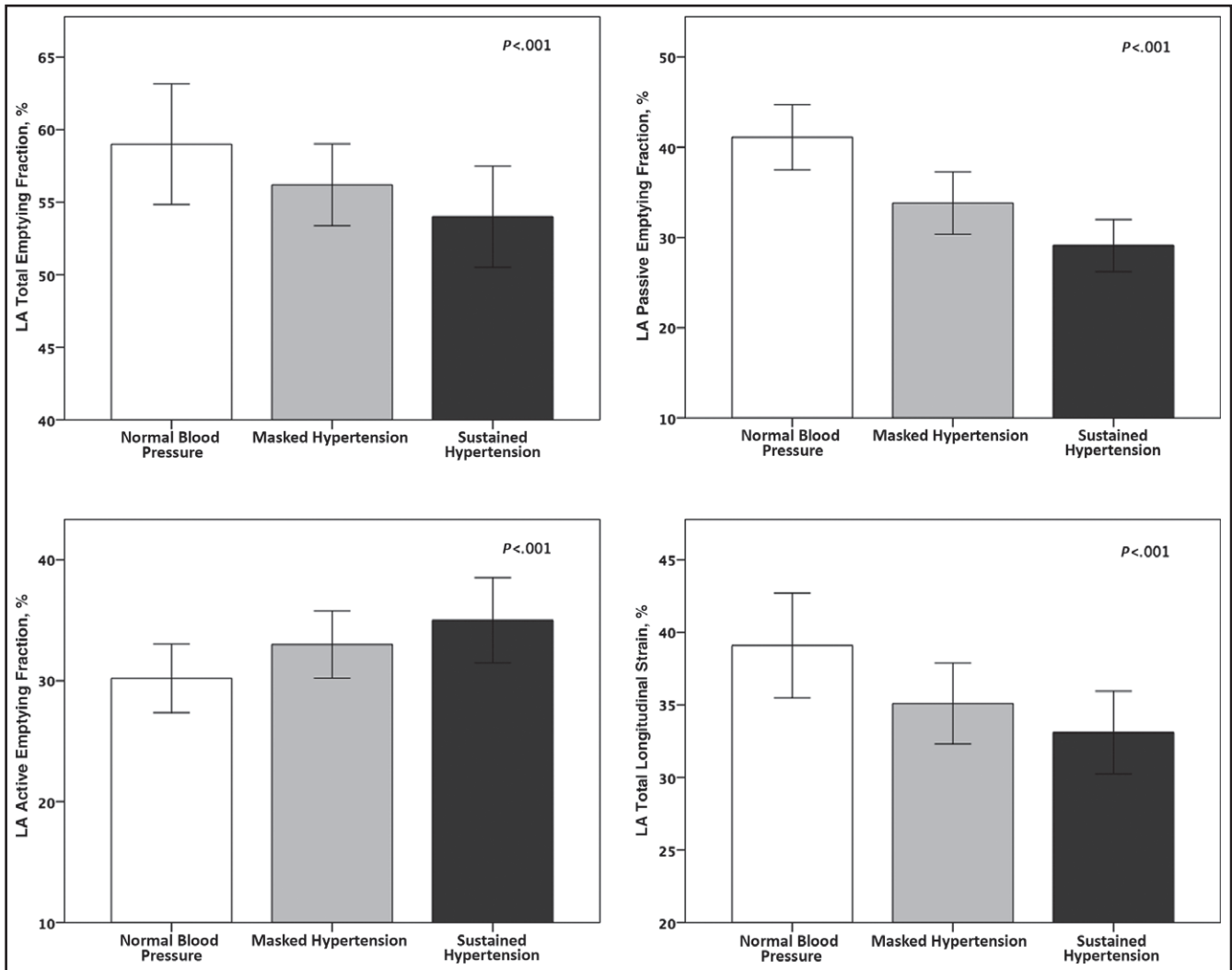


FIGURE. Left atrial (LA) phasic function assessed by echocardiographic volumetric method and LA total longitudinal strain.

longer in our population. Furthermore, one should consider that afterload is permanently increased in sustained hypertensive patients, which could also significantly contribute to LA dysfunction. This is not the case with MH.

Interestingly, there are also authors who did not find any significant difference in LA function between normotensive and hypertensive patients using volumetric methods.²⁰ Nevertheless, two-dimensional speckle tracking enabled the detection of difference in LA conduit and pump function between the observed groups.²⁰ This could indicate better sensitivity of strain method over volumetric method in the evaluation of LA phasic function. Our analysis showed significant matching between strain and volumetric analysis of LA phasic function, which could be important from a clinical aspect. The comprehensive volumetric evaluation of LA function could be challenging and time demanding as a result of the assessment of three LAVs and calculation of

all other parameters. This is why the volumetric LA assessment could be difficult in everyday clinical practice. On the other hand, LA strain analysis is rapid and provides the same quality of results, which makes this method more attractive for clinicians.

LA enlargement is associated with increased cardiovascular morbidity, primarily atrial fibrillation and stroke, and all-cause mortality in the general population.^{11,12} Furthermore, LA dysfunction is related to worsening of LV diastolic function and development of hypertensive heart failure, as well as with atrial fibrillation occurrence,^{21–24} which could partially explain the higher cardiovascular morbidity and mortality rates in MH individuals. Oki and colleagues²⁴ showed that hypertension plays an important role in the occurrence of heart failure with preserved EF due to impaired LA–LV–arterial coupling. Using cardiac magnetic resonance imaging, investigators showed that LA phasic dysfunction could provide important information about LA

fibrosis, which is one the most important pathophysiological substrates for atrial fibrillation development.²⁵

Our study also showed a significant increase in LA stiffness in MH patients, which represents one more risk factor for occurrence or worsening of LV diastolic function and atrial fibrillation.^{26,27} LA stiffness could be related to increased arterial stiffness in hypertensive patients.²⁸ In addition, impairment of LV diastolic function, as well as atrial fibrillation, induces diastolic heart failure that further increases LV filling pressure and causes deterioration of LV diastolic function that cannot be compensated with elevated LA pump function, which declines and additionally worsens LA and LV functions. It is of substantial clinical importance to detect the beginning of this cascade in MH and hypertensive patients before “circle vicious” occurs.

The hemodynamic and structural changes in MH could be explained by overactivity of biohumoral systems such as the adrenergic nervous system and existing insulin resistance.²⁹ Grassi and colleagues²⁹ demonstrated that MH was characterized by a marked sympathetic overdrive, which could also explain insulin resistance (increased glucose and insulin levels, as well as Homeostasis Model Assessment index) in this population. Sympathetic overactivity is associated with structural LA remodeling that further induces LA functional and mechanical changes in MH patients. LA changes could partly be explained by increased BP values, but also by increased LVMI in MH and sustained hypertensive individuals.³⁰

LIMITATIONS

The current study has several limitations. This research involved a limited number of patients. Nevertheless, we demonstrated statistically important differences between various observed groups. Echocardiographic assessment of LA function and mechanics could be significantly affected by the quality of ultrasound images. The authors used different definitions for MH, which could potentially restrict the comparison with other investigations. A causal relationship between MH and LA phasic function could not be demonstrated because of the cross-sectional nature of our study.

CONCLUSIONS

This investigation indicated the significant impact of MH on LA functional and mechanical remodeling. Conduit and reservoir LA functions were significantly reduced in MH and sustained hypertensive patients, whereas booster pump LA function was compensatory amplified in these patients. LA stiffness was significantly increased in MH patients, which could contribute to deterioration of LV diastolic function in this group. Impaired LA function and mechanics could potentially explain higher cardiovascular morbidity and mortality in MH individuals. However, further follow-up studies are essential for estimation of the potential predictive value of LA remodeling in patients with MH.

Conflict of Interest: None.

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